

A scatter plot showing the relationship between mean daily turnover (log scale) and return minus turnover. The x-axis is labeled 'mean daily turnover, log scale (fraction of the portfolio traded)' and ranges from 10^{-2} to 10^0 . The y-axis is labeled 'return_minus_turnover'. A dashed horizontal line is drawn at approximately y = 0. The plot compares three groups: 'human canon' (grey circles), 'optimiser' (orange circles), and 'LLM designer' (blue circles). The LLM designer points are clustered at low turnover (around 10^{-2} to 10^{-1}) and high return minus turnover. The optimiser points are scattered across the turnover range, generally showing a downward trend. The human canon points are clustered at high turnover (around 10^0) and low return minus turnover.

Group	mean daily turnover (log scale)	return_minus_turnover (approx.)
LLM designer	1.2×10^{-2}	0.95
LLM designer	2.5×10^{-2}	0.95
LLM designer	3.5×10^{-2}	0.85
LLM designer	3.5×10^{-2}	0.75
optimiser	7.5×10^{-2}	0.65
optimiser	1.2×10^{-1}	0.60
optimiser	2.5×10^{-1}	0.35
human canon	8.5×10^{-1}	0.05
human canon	9.0×10^{-1}	0.05
human canon	9.5×10^{-1}	0.05
human canon	9.5×10^{-1}	0.02
human canon	9.8×10^{-1}	0.02

Horizontal bar chart showing Sharpe ratios for various strategies. The x-axis is labeled 'Sharpe' and ranges from -0.4 to 1.2. The y-axis lists 17 strategies. Bars are color-coded: grey for negative Sharpe, orange for positive but below 0.85, and blue for positive and above 0.85. Dashed lines represent 'GROSS, before costs' and solid lines represent 'NET, as scored'.

Strategy	NET, as scored (Solid)	GROSS, before costs (Dashed)
mean_variance_utility	-0.30	0.82
raw_return	-0.30	0.82
volatility_scaled_return	-0.22	0.87
return_minus_variance	-0.22	0.92
log_growth	-0.20	0.93
return_minus_cvar	-0.32	0.93
return_minus_downside	-0.20	0.97
return_minus_drawdown	-0.20	1.03
differential_sharpe	-0.20	1.05
cma_es	0.44	1.06
random_search	0.92	1.09
differential_downside_ratio	-0.16	1.09
tpe	0.89	1.12
bayes_opt	1.05	1.12
placebo_shuffled	1.11	1.17
placebo	1.15	1.17
return_minus_turnover	1.16	1.17
scalar_cvar5	1.16	1.17